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NON-LINEAR TIME-TRANSIENT ROTOR DYNAMIC ANALYSES OF GEARED SYSTEMS

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ABSTRACT

The ability to accurately predict the response of rotating machinery to external forces and to assess system-level stability for different modes is crucial from a reliability and preventive maintenance perspective. Geared systems, in particular, contain many complexities, which may lead to instability and even chaotic vibration behavior. No methods for determining the effects of the dynamic meshing forces on the vibrations of complete shaft/bearing systems have been proposed in the literature. Several time-transient and steady-state models for analyzing gear forces and deflections have been proposed, but they focus primarily on the dynamics of the gearbox itself and neglect vibration transmission through the remainder of the drive-train. Models that do incorporate other components of the drive-train propose simplified lumped-parameter models for the shafts and bearings. Recent models have used the finite element method to couple the lateral, torsional, and axial degrees-of-freedom of geared shaft systems to the forces and moments exchanged between the gears via stiffness matrices. Other models in literature capture the backlash non-linearity and the state-varying mesh stiffness and observe the time-transient response of the gearbox and simplified shaft/bearing structure. A finite element formulation of complete geared systems, which couples the axial, lateral, and torsional degrees-of-freedom, is developed in which the shaft is modeled with Timoshenko beam elements and captures the forces and moments due to gyroscopic effects, and rotational accelerations due to start-up. It includes the capability of modeling non-linear contact loss due to backlash clearance and parametric excitations resulting from the state-varying mesh stiffness and solves the time-transient state

equations for the displacements and velocities of the shafts using the direct Runge-Kutta method.

INTRODUCTION

The concept of gearing and the benefits of its mechanical advantage have been known since the beginnings of rotating machinery and yet many aspects of their dynamics remain to be understood. Gearbox dynamics are important to understand because of their prevalence in a wide variety of rotating machinery and because they can substantially transmit and excite vibration throughout the entire structure. Automobile transmissions, jet engines, turbine-generator systems are just a few of many applications that benefit from the inherent torque-speed mechanical advantages.

Despite these advantages, geared systems are susceptible to failure via dynamic tooth stresses, pitting and scoring, and self-exciting instabilities [1-4]. Noise radiation has also been a concern particularly for naval applications that require stealth. Models of varying complexity and detail have been necessary to explore each of these phenomena and range from SDOF torsional models to detailed FEA models of the gear mesh interface [1]. Despite the increases in the complexity of tooth compliance models using FEA, the results showed little difference from those of the pioneering mass-spring models with the exception of high-speed cases [1]. Researchers determined, however, that more general models that incorporate the flexibility of other machine components were necessary for several practical applications. Vibration coupling between the gears and their respective shafts and bearings could no longer be neglected when they have comparable stiffness. Some models included lateral and torsional flexibility of the shafts and gear

teeth [5-7], while others focused exclusively on the torsional vibration of the shafts and gears [8-11] and still other torsional models of the shaft with rigid gear teeth [12,13].

More recent studies have investigated the effects of the state-varying mesh stiffness and non-linear backlash phenomena on the vibration of geared shafts [14-19]. Others have explored the influence of non-linearities not only from the gear mesh but from other components such as bearing clearance, oil film, and flexible supports [20-22]. Some instability studies have extended the inclusion of state-varying stiffness and non-linear backlash to multi-stage geared systems [23-26]. Parker et al. used a semi-analytical finite element formulation combined with a contact mechanics model to evaluate the mesh stiffness at each time step over a single tooth pass and compared the vibration results with some SDOF models to experimental data [27]. The results indicated high sensitivity to the spectral content of the rectangular waves being used to represent the state-varying mesh stiffness. Bifurcation and transitions to chaos have been studied using SDOF models [28-31].

Despite the advances in literature of modeling the effects of state-varying mesh stiffness and the backlash non-linearity, the previous studies have focused exclusively on either SDOF systems or simplified lumped parameter models for the geared shafts. Since the mass of the shafts are typically ignored and the shaft stiffness is approximated by single linear springs, significant details for evaluating machine safety, such as rotor whirl orbits, are inherently oversimplified. Detailed knowledge of the rotor whirl orbits are not only essential to recognizing potential for high-amplitude vibration but also provide the analyst with insights that may correct the problem. These necessary insights may only be discovered by discretizing the shaft into finite elements while exploring the rotor dynamic behavior due to the presence of gear mesh dynamic forces. Modern computing power is robust enough to use beam finite elements for time-transient calculations within a reasonable amount of time.

In addition, these studies have neglected to use excitations outside of external torques and mesh frequency excitations at the gear mesh interface, while conventional rotor dynamic analyses for high-speed machinery typically require the inclusion of lateral excitations such as unbalance. Furthermore, gyroscopic moments play a crucial role in predicting the forward and backward whirl modes of rotors and these have also been omitted in all previous studies.

A finite element approach to studying the time-transient behavior of generalized parallel-shaft geared systems is proposed in this study. The axial, lateral, and torsional DOFs of the geared shafts are included and coupled via transformation matrices that relate the generalized displacements and forces along the line of action (LOA) to those of the shaft nodes [32]. State-varying mesh stiffness is incorporated and contributes to parametric excitations at the gear mesh interface. Gear tooth backlash is modeled as a piecewise-linear function that is dependent on the dynamic transmission error (DTE) at each time step. Rotational accelerations are accounted for in the equations of motion and are applicable to analyzing start-up or wind-down

conditions. The generalized displacements and velocities of each node are evaluated at each time step via the direct Runge-Kutta method. A summary of the capabilities of other models relative to the proposed approach is shown in Table 1.

Table 1. Summary of previous geared-system models.

Parameters	[14]	[20]	[24]	[27]	[21]
Model Type	SDOF	SDOF	Lump	FEA	Lump
Backlash	Yes	Yes	No	Yes	No
Mesh Vary	Yes	Yes	Yes	Yes	Yes
Shaft Flex	No	No	Yes	No	Yes
Gyroscopics	No	No	No	No	No
Multi-Stage	No	No	Yes	No	No
Unbalance	No	No	No	No	Yes
Lat/Tors Coupling	No	No	No	No	No

The primary objective of this study is to extend the existing modeling for parametric excitations due to state-varying mesh stiffness and non-linearities such as tooth backlash clearance to more realistic geared shaft bearing systems. While commercial FEA codes for geared systems exist, they focus on the design of the gearbox and all of its complexities such as the housing so that it may integrate efficiently with the expected loading conditions from the remainder of the drive-train. 3-D finite element models are used to evaluate the deflections of the shafts, bearings, gear teeth and bodies, and housing for both steady-state and time-transient conditions. In addition, detailed dynamic analyses of the mesh interface reveal the transmission error excitations over a single-tooth pass. Despite the complexity of these models, it is generally understood that the use of 3-D finite elements may require vast computing power and may have substantially longer solve times especially for transient analyses. Most importantly, the emphasis for these analyses is still, nevertheless, focused on the design of the gearbox for mitigating noise, but is not directed towards consideration of the potential whirling instabilities that may be induced for the rest of the drive-train. Such a subtle difference in modeling considerations could easily affect the conservativeness in determining a geared system's stability.

There is currently interest in the rotor dynamics community in understanding the effects of different gear parameters on the whirling characteristics of realistic geared shaft systems. Although detailed methods have been presented in literature and commercial software is available, the results are rendered impractical to industry when their emphasis is on the dynamics of the gearbox rather than on the dynamics of the complete machine. This study is expected to promote discussion of the many parameters involved in the modeling of state-varying mesh stiffness and backlash and how they can influence the transient behavior of complete geared systems.

NOMENCLATURE

a, b – Fourier series coefficients
 b_s – Backlash clearance, mm

$\mathbf{C}$ – Damping matrix
 c – Gear contact ratio
 E – Elastic Modulus, Pa
 $\mathbf{f}$ – External force vector
 $\mathbf{G}$ – Gyroscopic matrix
 $\mathbf{K}$ – Stiffness matrix
 K – Gear mesh stiffness, N/m
 K_a – 0-P amplitude of state-varying stiffness, N/m
 K_g – Average gear mesh stiffness, N/m
 K_v – Gear mesh state-varying stiffness, N/m
 $\mathbf{M}$ – Inertia matrix
 N – Number of gear teeth
 p – Gear mesh phase, rad
 r – Gear base circle radius, m
 s – Fourier coefficient index
 t – Time, secs
 $\mathbf{u}$ – Displacement vector
 w – Gear tooth face width, mm
 α_n – Gear tooth normal pressure angle, rad
 β – Gear tooth helical angle, rad
 δ – Dynamic transmission error (DTE), mm-rad
 ϕ – Gear direction cosine, rad
 φ – Gear orientation angle, rad
 ω_d – Damped natural frequency, rad/sec, rpm, Hz
 Ω – Rotational speed or excitation frequency, rad/sec

Subscripts

i – Node number
 j – Node number
 x – Horizontal displacement
 y – Vertical displacement
 z – Axial displacement

METHODS

Beam elements are widely used to model rotors as they have been shown to produce accurate results when compared to experimental data [33]. The methods have applications towards performing steady-state and time-transient rotor dynamic analyses and have lateral, torsional, and axial DOFs. Solving the rotor dynamic equations of motion with coupled degrees of freedom are necessary and will produce more accurate results than solving the non-coupled ones because some modes exhibit combined lateral and torsional displacements.

The approach models the effects of gears and gear mesh stiffness on the vibration characteristics of rotating machines using the finite element method. The gears are treated as a pair of rigid lumped masses and inertias that influence the mass, gyroscopic, and stiffness properties of the corresponding shaft nodes in the finite element matrices. The forced response rotor dynamic equations of motion for linear steady-state conditions can be represented by the following matrix equation.

$$\mathbf{M}\ddot{\mathbf{u}} + (\mathbf{C} + \Omega\mathbf{G})\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = \mathbf{f} \quad (1)$$

In a linear rotor dynamic analysis, the effects of the gears and the gear mesh will significantly contribute to the global mass, speed, gyroscopic, and stiffness matrices only. The gear mesh lubricant stiffness and damping effects are neglected.

The mass and gyroscopic contributions to the global matrices are treated the same as those of lumped disks at the nodes of the gear and pinion in rotor dynamics models and therefore will not be emphasized. The gear mesh effective lateral, torsional, and axial stiffness is modeled as a 12x12 matrix that relates the gear mesh forces and moments, or generalized forces, with each of their respective degrees of freedom. This gear mesh finite element consists of two nodes, i and j , respectively as shown in Figure 1, where the nodes designate the location of the gear or pinion on the connecting shafts. Each node has six degrees of freedom, which consists of three translations and three angular displacements. The incorporation of this axially, laterally, and torsionally coupled mesh stiffness finite element will provide more accurate displacement solutions for a geared system in a free vibration or forced response rotor dynamic analysis.

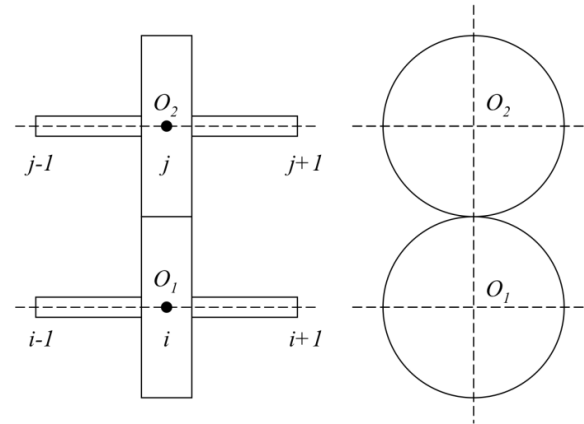


Figure 1. Gear mesh finite element representation [34]

The generalized displacements are shown in the displacement vector, labeled as $\mathbf{u}$, and contain those corresponding to the shaft center of one gear at node i and the other gear at node j , as indicated by the subscripts.

$$\mathbf{u}_n = [x_n \ y_n \ z_n \ \theta_{xn} \ \theta_{yn} \ \theta_{zn}]^T \ (n = i, j) \quad (2)$$

Correspondingly, the generalized forces acting on both nodes may be represented by the external force vector, labeled as $\mathbf{f}$.

$$\mathbf{f}_n = [F_{xi} \ F_{yi} \ F_{zi} \ M_{xi} \ M_{yi} \ M_{zi}]^T \ (n = i, j) \quad (3)$$

These generalized forces are incorporated into the rotor dynamic model through a shift to the left-hand side of the equations of motion since the generalized forces are treated as linear with respect to the generalized displacements. This equivalent stiffness matrix, $\mathbf{K}_{mesh}$, relates the generalized forces

with the generalized displacements through the following matrix equation.

$$\begin{bmatrix} f_i \\ f_j \end{bmatrix} = -[K_{mesh}] \begin{bmatrix} u_i \\ u_j \end{bmatrix} \quad (4)$$

LINEAR TIME-INVARIANT MESH STIFFNESS MATRIX

The element stiffness matrix K_{mesh} is defined below, where K_g is the average gear mesh stiffness, and K_{ii} , K_{ij} , K_{ji} , and K_{jj} are 6x6 sub-matrices that account for the coordinate transformations from the pitch point of the gear mesh to the global coordinate system of the shaft centers.

$$K_{mesh} = K_g \begin{bmatrix} K_{ii} & K_{ij} \\ K_{ji} & K_{jj} \end{bmatrix} = K_g \begin{bmatrix} K_{ii} & K_{ji}^T \\ K_{ji} & K_{jj} \end{bmatrix} \quad (5)$$

A crucial parameter to the element stiffness matrix is the average gear mesh stiffness, K_g , which accounts for the tooth compliance along the path of force transmission, called the LOA. It is assumed that the stiffness of the rest of the gear body will be much more rigid than that of the teeth, which suggests that the tooth stiffness will dominate the gear dynamics. The meshing stiffness between a single pair of teeth is found either from experimental data or from previously reported analytical formulas. The analytical formula used to compute single tooth-tooth contact stiffness is provided in Equation (6) [35].

$$K_g = \frac{wE_iE_j}{9(E_i + E_j)} \quad (6)$$

Where w is the tooth face width, and E_i and E_j are the gear and pinion elastic moduli.

The mesh stiffness formula, however, only represents the stiffness of a single pair of teeth in contact. As the gears rotate, a state-varying mesh stiffness develops because the number of pairs of teeth in contact, also known as the contact ratio, alternate typically between one and two throughout the mesh cycle. Since most gear manufacturers provide the average contact ratio, Equation (6) may be modified as the following.

$$K_g = \frac{cwE_iE_j}{9(E_i + E_j)} \quad (7)$$

Where c is the average contact ratio as specified by the gear manufacturer.

The transformation matrix is generalized to account for the macro geometry of both spur and helical geared systems. These geometric parameters, which include normal pressure angle, helical angle, pitch radii, and orientation angle, are accounted for when relating the displacement of gear teeth along

the LOA to its components in the coordinate system of the shaft center [34]. The displacements at the pitch point are resolved into the components of the shaft coordinate system through two coordinate transformations. The first transformation makes use of the normal pressure angle and helical angle to resolve components along the LOA into normal and tangential components along the pitch circle. The second transformation makes use of the shaft orientation angle to relate the components along the pitch circle to those of the shaft center.

Spur geared systems require only three degrees of freedom per node because the forces and moments act solely in the plane of rotation, or in the X-Y plane in the diagram below. The displacements of interest for each gear node would be x , y , and θ_z , and therefore $\mathbf{u}$ would consist of only six unknown displacements. That is, three displacements for the pinion and three displacements for the gear.

Helical geared systems, however, as shown in Figure 2, require six degrees of freedom per node because the forces and moments now act in 3-D space and must also be functions of the helical angle, β . Therefore, $\mathbf{u}$ must include all 12 generalized displacements if we are to include those of the gear and pinion.

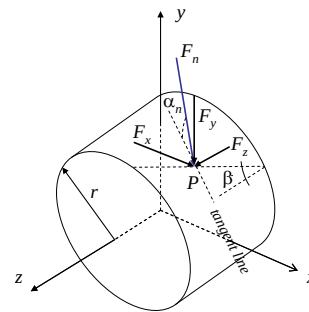


Figure 2. Gear Forces and Parameters [34]

The equations of motion that relate the generalized forces to the generalized displacements were obtained by applying a force balance along the LOA. The transmitted force is proportional to the net displacements of the teeth along the LOA through a component of the mesh stiffness matrix. The transmitted force and displacement along the LOA can be resolved into components of the shaft center coordinate system through the two coordinate transformations involving the parameters mentioned above. Each element of the gear mesh stiffness matrix may then be evaluated by varying individual generalized displacements and determining the resulting generalized forces required to produce that individual deflection. The direction cosines are used to resolve the transmitted force into components along the pitch circle and are convenient for notational purposes. They are functions of the helical angle and normal pressure angle as shown by Equation (8).

$$\begin{aligned} \cos \phi_x &= \cos \beta \cos \alpha_n \\ \cos \phi_y &= \sin \alpha_n \\ \cos \phi_z &= \sin \beta \sin \alpha_n \end{aligned} \quad (8)$$

K_{ii} , K_{ij} , K_{ji} , and K_{jj} are 6x6 sub-matrices that represent the geometric contributions of the helical angle, normal pressure angle, pitch radii, and the orientation angle of the gears to the element stiffness matrix. Those sub-matrices are expressed in Equation (16) in Annex A.

After applying the coordinate transformations and multiplying through by the average gear mesh stiffness, we have obtained an element stiffness matrix that represents a linearized time-invariant relationship between the generalized forces exchanged between the gears, and the generalized displacements at the corresponding shaft locations. This method may be applied to both spur and helical gears and illustrates the contribution gear dynamics provide to the deformation and vibration of axial, lateral, and torsionally coupled rotor dynamic systems with constant gear mesh stiffness.

LINEAR STATE-VARYING MESH STIFFNESS

The derivation for the mesh stiffness, presented in the previous section, was a substantial simplification of the dynamic meshing forces exchanged between the gears. A closer approximation should include an oscillating state-varying function that is offset by the average mesh stiffness since the number of teeth in contact change as the teeth transition in and out of mesh. The equivalent meshing stiffness along the LOA then becomes.

$$K = K_g + K_v(\Omega, t) \quad (9)$$

Where K_g is the average mesh stiffness and $K_v(t)$ is the state-varying contribution. Several authors have approximated the state-varying mesh stiffness as a set of rectangular waves via Fourier series [14, 20, 23].

$$K_v(t) = 2K_a \sum_{s=1}^{\infty} (a^{(s)} \sin(sN_i\Omega_i t) + b^{(s)} \cos(sN_i\Omega_i t)) \quad (10)$$

This formulation of the gear mesh stiffness allows for additional studies on gear parameters. These include the gear mesh frequency, contact ratio, relative magnitude of the state-varying stiffness, and the phase lead/lag between multiple meshes. The rotational speed and number of gear teeth dictate the mesh frequency, where N is the number of gear teeth and Ω is its corresponding rotational speed in radians per second.

The series coefficients are defined by the following equations, which are dependent upon the contact ratio, c , and the mesh phasing, p .

$$\begin{aligned} a^{(s)} &= -\frac{2}{s\pi} \sin[s\pi(c - 2p)] \sin(s\pi c) \\ b^{(s)} &= -\frac{2}{s\pi} \cos[s\pi(c - 2p)] \sin(s\pi c) \end{aligned} \quad (11)$$

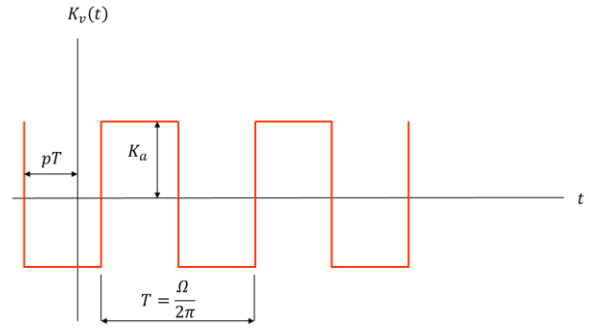


Figure 4. Rectangular wave form representation of state-varying mesh stiffness [24].

The first three or four terms of the Fourier series have been sufficient approximations to the rectangular wave form, which may produce greater accuracy in the vibration predictions since the true mesh stiffness changes more gradually and contains oscillatory imperfections [24].

The contact ratio indicates the average number of teeth in contact upon following a single tooth pass in to and out of mesh. Figure 5 depicts the rectangular wave forms of the Fourier series over a range of contact ratios between 1.25 and 2.0 at a mesh frequency of 25 radians per second expanded to 25 terms. The average mesh stiffness was set to 100 and the 0-P amplitude of oscillation is 1. A contact ratio of 1.5 dictates that two pairs of gear teeth will be in contact for the same duration as just one pair. Integer values stipulate that there will be no oscillation in the gear mesh stiffness because there is perfect overlap as teeth move into and out of engagement.

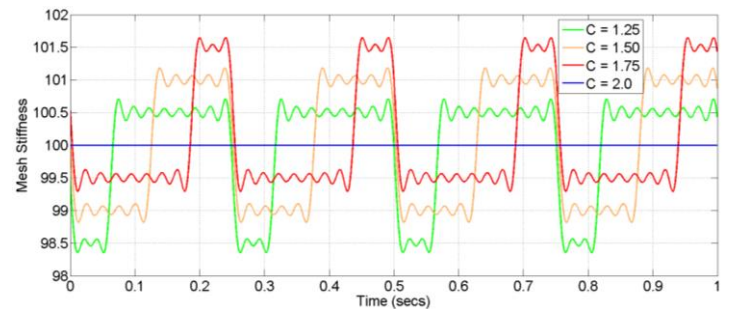


Figure 5. 25 Term Fourier Series Expansion of Mesh Stiffness.

Another parameter worth investigating would be the sensitivity of the number of Fourier expansions used in obtaining the forced response of the geared rotor system. The additional minor oscillations that are more visible with fewer Fourier series terms will induce excitations in the forced response analysis and it is questionable as to whether these more accurately reflect the physics of real gear meshes or whether they should be studied as undesirable computational “noise.” However, using higher order terms to yield closer approximations to the rectangular waveform could excite poorly damped modes at high resonant frequencies.

NON-LINEAR STATE-VARYING MESH STIFFNESS

The previous two sections considered the gear mesh forces as being directly proportional to the gear tooth displacements and rotations. Gear manufacturers design the teeth to mesh so that there exists a clearance region to provide space for oil film development, thermal expansion, and to account for manufacturing tolerances. The clearance region is formerly referred to as backlash and is the dominant non-linear factor in gear dynamics. This section considers contact loss in the gear teeth due to sufficient tangential displacement differences between the meshing gear and pinion. The dynamic transmission error (DTE) defines the differences between gear i and pinion j tooth tangential displacements and is represented in Equation (12).

$$\delta(t) = r_i \theta_i(t) - r_j \theta_j(t) \quad (12)$$

Where r_i and r_j represent the base circle radii for gear i and pinion j . If δ exceeds the backlash clearance, then tooth contact exists and the gear forces are treated as linearly proportional to the displacements as defined previously. When δ is within the backlash clearance, the gear forces must be zero because there is no tooth contact. The piecewise linear expressions for the gear contact force are shown below.

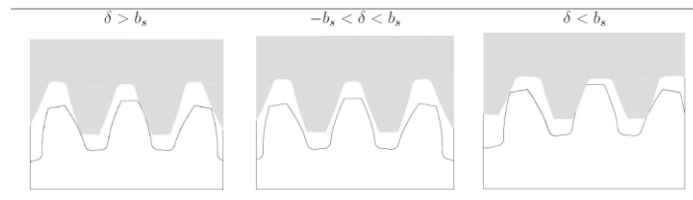


Figure 6. Gear tooth backlash clearance conditions in terms of DTE [16].

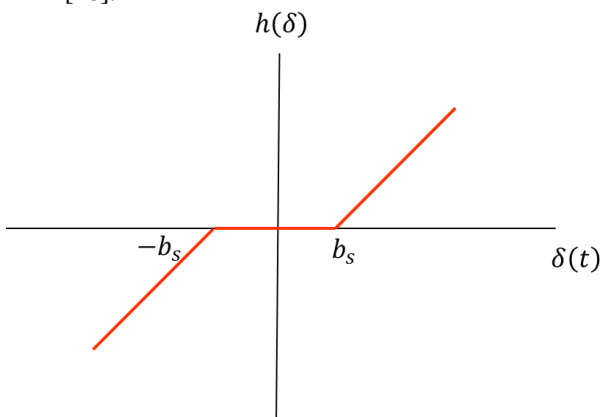


Figure 7. Contact loss criteria.

$$h(\delta) = \begin{cases} \delta - b_s, & \text{if } \delta > b_s \\ 0, & \text{if } |\delta| < b_s \\ b_s - \delta, & \text{if } \delta < -b_s \end{cases} \quad (13)$$

The dynamic meshing forces may now be summarized from the contributions of the static and state-varying mesh stiffness, the non-linear backlash function, transformation matrices, and the generalized displacements of the gear teeth.

$$\begin{bmatrix} f_i \\ f_j \end{bmatrix} = - \left[(K_g + K_v(\Omega, t)) h(\theta_i, \theta_j) \begin{bmatrix} [K_{ii}] & [K_{ji}]^T \\ [K_{ji}] & [K_{jj}] \end{bmatrix} \right] \begin{bmatrix} u_i \\ u_j \end{bmatrix} \quad (14)$$

The methods presented thus far may be used to evaluate the dynamic meshing forces at the gear teeth and to translate them into equivalent forces exchanged between the connected shafts. Computing the resulting non-linear gear mesh forces and applying it to a finite element model of a shaft system results in a new set of displacements and velocities at a future time step. These new sets of displacements and velocities subsequently produce a new set of dynamic forces that are applied to the geared rotor bearing model. The non-linear state-varying equations of motion may be expressed in Equation (15) and include unbalance forces with acceleration terms.

$$\begin{aligned} M\ddot{u}(t) + [C + \Omega(t)G]\dot{u}(t) + [K + \dot{\Omega}(t)G]u(t) \\ = F_n(u, \dot{u}, t) + F_{unb}(\Omega, \dot{\Omega}, t) \end{aligned} \quad (15)$$

RESULTS

The methods shown thus far are applied to a two-stage gear model presented by Walha et al. [23]. Their model consists of three flexible and massless shafts and two identical gear stages for a total of 12 DOFs. The motor shaft is driven by a constant torque and the system accelerates from rest. Two case studies were presented in their paper that contrast the tooth velocity fluctuations resulting from backlash and state-varying stiffness present in either one or both gear stages. Details of the gear and shaft models are provided below in Table 2 and indicate that the focus of the analyses was on torsional deformation since the shaft lateral stiffness was omitted.

Table 2. Original shaft and gear mesh parameters [23].

Shaft Material Density (kg/m ³): 42CrMo4	7860
Motor Torque (N-m)	100
Bearing Stiffness (N/m)	1e7
Torsional Shaft Stiffness (N-m/rad)	1e5
Pressure Angle (degrees)	20; 20
Number of Teeth	30:45; 30:45
Average Mesh Stiffness (N/m)	3e7; 3e7
Contact Ratio	1.592; 1.592

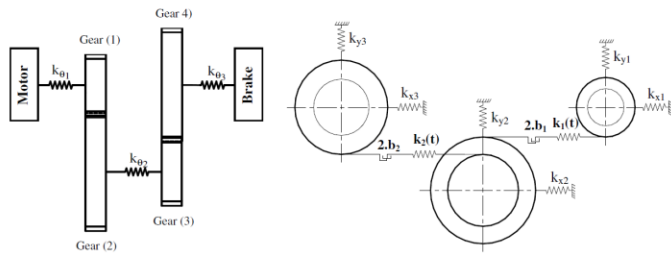


Figure 8. Lumped parameter model of a two-stage geared system [23].

A more detailed representation of the model with lateral-torsional coupling at the gear mesh interface is presented in this paper. Each shaft has been divided into 20 elements with a length and outer diameter to produce the equivalent torsional stiffness of the original model. Table 3 summarizes some additional model details that dictate inertial and lateral properties of the shafts and is illustrated in Figure 9.

Table 3. Extended parameters to include shaft inertia and lateral properties.

Shaft Length (m)	1.02
Shaft Outer Diameter (m)	0.06
Bearing Span (m)	0.81
Shaft Bending Stiffness (N/m)	4.41e6
Bearing Damping (N-sec/m)	1751
Shaft Mass (kg)	22.53

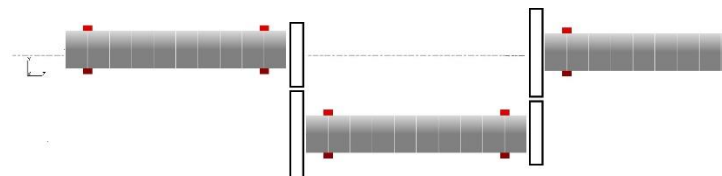


Figure 9. Augmented model of geared system with lateral and torsional shaft properties defined.

A summary of the first ten natural frequencies and mode shapes of the geared system with constant gear mesh stiffness are provided in Table 4. Figure 10 depicts the 6th mode shape and is emphasized because of its prevalence in the system transient response. Note that the backward and forward modes have different natural frequencies due to gyroscopic moments and the presence of combined lateral and torsional mode shapes.

Table 4. Natural frequencies and mode shapes of geared system with constant gear mesh stiffness.

Mode	ω_d (RPM)	Log Dec	Lateral	Torsional
1	7294	0.292	Backward	X
2	7399	0.267	Backward	N/A
3	7408	0.264	Backward	N/A
4	7413	0.265	Forward	N/A

5	7414	0.266	Forward	N/A
6	7418	0.266	Forward	N/A
7	10948	0.500	Backward	X
8	11340	0.642	Backward	X
9	12273	0.706	Backward	N/A
10	12287	0.706	Forward	N/A

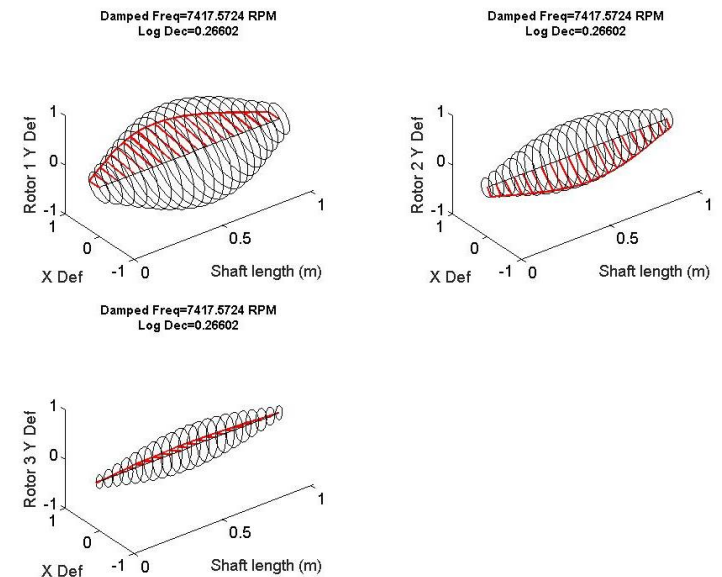


Figure 10. 6th mode shape of geared system.

As documented in the original model, a constant torque is applied to the motor shaft and all shafts subsequently accelerate from rest. An unbalance of 72 g-mm is placed on the central node of all three shafts to generate a response that would be consistent with the 6th mode. Two analyses are performed in this paper. The first sets the backlash clearance to 0 and has no state-varying mesh stiffness. Figure 11 depicts the transient whirling orbit response of the central node of the motor shaft. After 0.1 seconds from start-up, the unbalance forces excite the first critical speed (modes 4-6 in Table 4) and the amplitude in the X and Y directions increases. As the rotors continue to increase in speed, the excitation frequency due to unbalance becomes farther from the critical speed and the whirling amplitude decreases. A plot of the lateral transient response of the entire motor shaft is shown in Figure 12 and is consistent with mode shape 6. Additional bending of the non-drive end of the shaft from centerline results from the gear forces exchanged in the first stage.

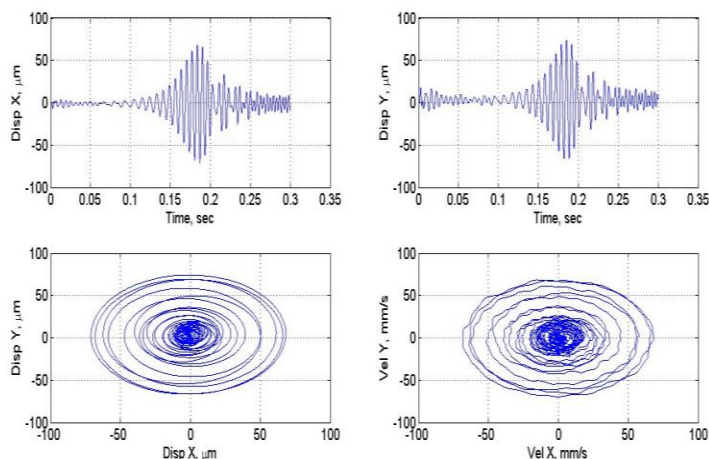


Figure 11. X and Y displacements and velocities of the central node of the motor shaft during start-up.

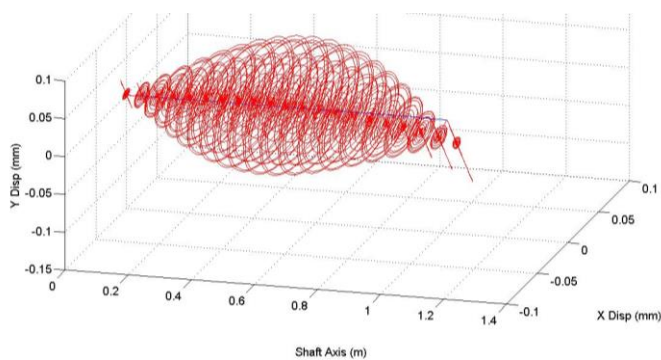


Figure 12. Lateral transient response of the motor shaft due to unbalance during start-up.

Responses of the other shafts may be summarized in Figure 13. X and Y displacements versus time and their respective Fast Fourier Transforms are presented. The motor shaft experiences a peak frequency around 8,400 RPM, which is above the damped natural frequency of mode 6. A likely reason for this discrepancy is that the shaft rotational speed is constantly increasing while its inertia causes a lag in the system response.

The idler shaft, however, experiences two peak frequencies. Since the idler shaft operates at a rotational speed that is 2/3 of the motor shaft rotational speed, its unbalance forcing function would not excite a natural frequency. The time-domain data suggests that at least one peak could result from the abrupt force and torque from the first and second gear stages. Such an arrangement of forces on both ends of the idler shaft could excite the 7th mode at 10,900 RPM as shown in Figure 14. Unbalance forces from the motor shaft may also contribute to the peak frequency in the idler shaft at 8,400 RPM due to the geared connection. The load shaft response amplitude is much smaller than the idler and motor shafts since it operated at 4/9 the rotational speed of the motor shaft and because the gear forces and torques are exerted on the drive end only.

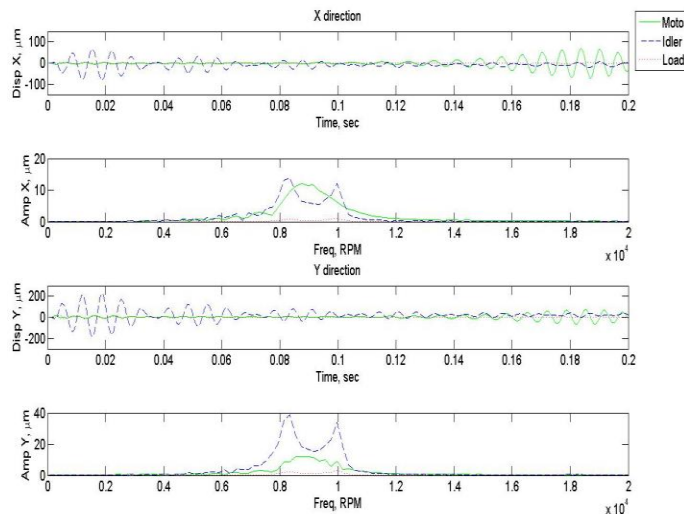


Figure 13. Time and frequency domain of the response of motor, idler, and load shafts.

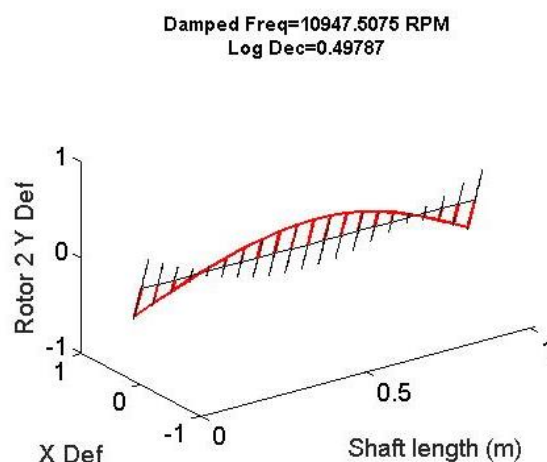


Figure 14. 7th mode shape of the idler shaft.

The second analysis is similar to the first except that a backlash clearance of 0.254 mm is specified for both gear stages and a range of state-varying mesh stiffness amplitudes between 0 and 50% of the average gear mesh stiffness are included. Figure 15 illustrates the results on the central node of the motor shaft, which suggest that the addition of state-varying mesh stiffness induces significant high-frequency vibration at both sudden onset and post-critical operating speed. Additional peaks in the FFTs become more apparent with state-varying stiffness amplitudes above 10% of the average gear mesh stiffness and they excite vibration at frequencies of 25,000, 83,000 and 97,000 RPM, which are beyond the first 10 modes recorded in Table 4.

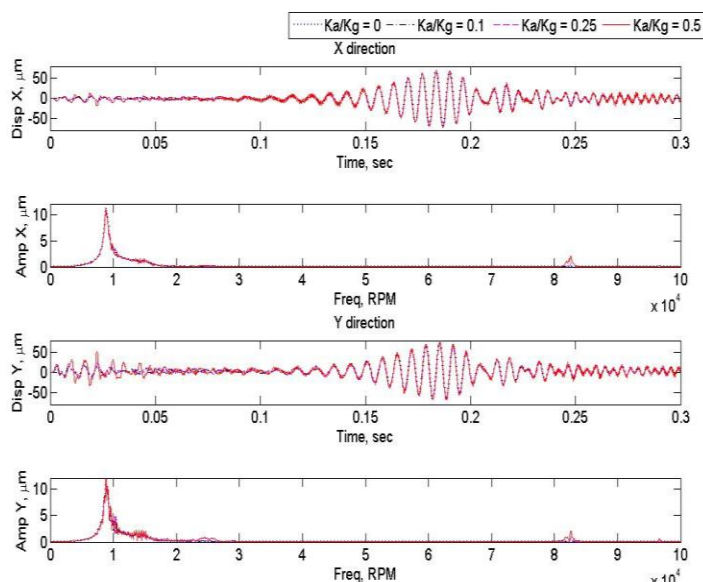


Figure 15. Time and frequency domain response of central node on motor shaft with different state-varying stiffness amplitudes.

The torsional component of vibration must also be examined since the gear mesh stiffness couples it with lateral vibration. The results, shown in Figures 16 and 17, indicate that the state-varying mesh stiffness contributes to substantial rotational speed fluctuations for both low and high frequencies. This could represent a component of instability since rotational speed fluctuations result in dynamic transmission error. Sufficient dynamic transmission error induces non-linearities such as tooth contact loss, which could excite other natural frequencies. Additional excitations would then promote further fluctuation in the rotational speeds of the geared shafts and the cycle would continue.

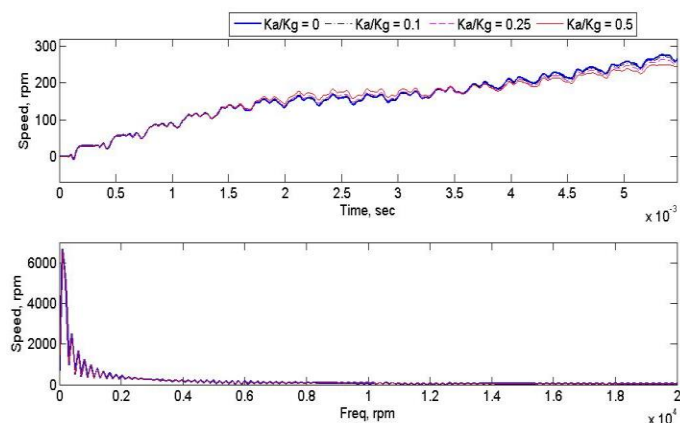


Figure 16. Rotational speed of the motor shaft in time and frequency domain under different state-varying stiffness amplitudes.

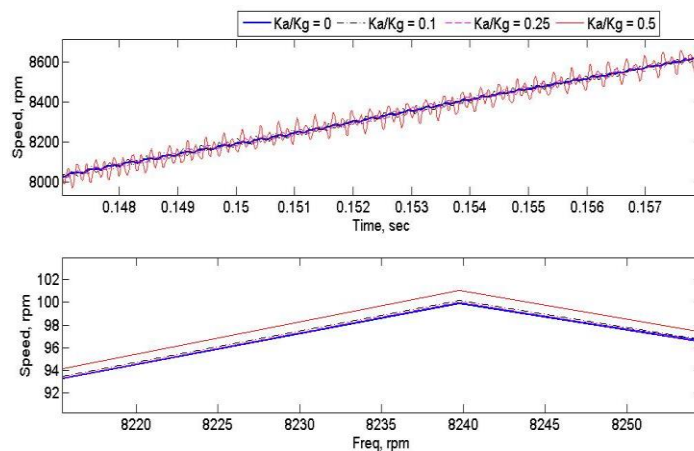


Figure 17. Zoomed in rotational speed of the motor shaft undergoing the first critical speed in time and frequency domain.

The methods presented have been applied to a two-stage spur gear model that is documented in the literature [23]. A constant torque was applied to the first shaft and the torsional and lateral displacements and velocities were observed for a specific unbalance configuration. Cases 1 and 2 differed by whether the gear stages had both backlash clearance and state-varying mesh stiffness or neither. The results indicated the importance of having a detailed model of the shaft dynamics because there is substantial feedback between the shaft displacements and the gear mesh dynamic forces. The inclusion of unbalance and the coupling of lateral and torsional vibration were necessary to illustrate this effect but are not present in previous literature.

CONCLUSIONS

Dynamic geared system analyses have extended into detailed investigations of non-linearities, such as backlash, and parametric excitations due to the state-varying mesh stiffness. Even though predictive tools for these forces are better known, they have yet to be applied to models that more accurately reflect realistic geared systems. While lumped parameter models provide great physical insights into studying these phenomena in gearboxes, their oversimplification of the physics and geometry of other system components, such as the rotors, renders them less useful to studying complete geared rotor-bearing systems and may produce misleading results. While commercial FEA codes for geared systems produce detailed models of the shafts and gears, they may require vast computational power to obtain a solution, especially for transient simulations. They are also written to help the manufacturer design a gearbox to integrate with the expected loading conditions from the rest of the machine, but this does not include potential whirling instabilities. The methods proposed in this study may greatly extend the modeling of dynamic meshing forces to more complex geared machinery and provide greater insights into geared system instability from a rotor dynamics perspective.

The scope of the current study is limited to exploring only a few parameters, but additional work that investigates the sensitivity of geared system response to the other parameters may be expected. Such parameters include unbalance magnitude, bearing damping, number of Fourier coefficients, mesh phasing, and contact ratio in conjunction with state-varying mesh stiffness and backlash clearance non-linearities under both start-up and steady-state operating conditions. Such options and discussion would produce predictive tools that would lead to a better understanding of proper modeling techniques for geared systems and therefore to an overall enhancement of geared system reliability and life.

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REFERENCES

- [1] Nevzat Özgüven, H., and Houser, D. R., 1988, "Mathematical Models used in Gear Dynamics—a Review," *Journal of Sound and Vibration*, 121(3) pp. 383-411.
- [2] Bollinger, J., and Harker, R., 1967, "Instability Potential of High Speed Gearing," *Journal of the Industrial Mathematics*, 17pp. 39-55.
- [3] Benton, M., and Seireg, A., 1981, "Factors Influencing Instability and Resonances in Geared Systems," *Journal of Mechanical Design*, 103(2) pp. 372-378.
- [4] Nataraj, C., and Whitman, A., 1997, "Parameter excitation effects in gear dynamics," *ASME Design Engineering Technical Conferences*, Anonymous.
- [5] Fukuma, H., Furukawa, T., and Aida, T., 1973, "Fundamental Research on Gear Noise and Vibration: (6th Report, Generation Mechanism of Radial and Axial Vibration of Spur Gears)," *Bulletin of JSME*, 16(97) pp. 1094-1107.
- [6] Rettig, H., 1976, "Vibrations in Gear Drives; Test Results and Calculation Method for Dynamic Tooth Forces."
- [7] Salzer, M. W., 1975, "The Dynamics of a Layshaft Gearbox," Ph.D. Thesis, Cambridge University.
- [8] Tordion, G. V., 1962, "The mechanical impedance approach to the dynamics (torsional vibrations) of geared systems," *American Gear Manufacturers Association Paper*, Anonymous Paper No. 209.04.
- [9] Wang, S., and Morse, I. E., 1972, "Torsional Response of a Gear Train System," *Journal of Manufacturing Science and Engineering*, 94(2) pp. 583-592.
- [10] Wang, S., 1974, "Analysis of Nonlinear Transient Motion of a Geared Torsional," *Journal of Manufacturing Science and Engineering*, 96(1) pp. 51-59.
- [11] Kisher, B., 1979, "Effect of gear errors on non-linear vibrations of a gear train system," *Proceedings of the International Federation of the Theory of Machines and Mechanisms Fifth World Congress*, Anonymous Montreal, Canada, pp. 1122-1125.
- [12] Kasuba, R., 1971, "Dynamic Loads on Spur Gear Teeth by Analog Computation," *American Society of Mechanical Engineers*, Paper No. 71-DE-26.
- [13] Lees, A. W., 1984, "Dynamic loads in gear teeth," *Proceedings of the Third International Conference on Vibrations in Rotating Machinery*, Anonymous Institution of Mechanical Engineers, pp. 73-79.
- [14] Theodossiades, S., and Natsiavas, S., 2000, "Non-Linear Dynamics of Gear-Pair Systems with Periodic Stiffness and Backlash," *Journal of Sound and Vibration*, 229(2) pp. 287-310.
- [15] Moradi, H., and Salarieh, H., 2012, "Analysis of Nonlinear Oscillations in Spur Gear Pairs with Approximated Modelling of Backlash Nonlinearity," *Mechanism and Machine Theory*, 51pp. 14-31.
- [16] Amezket, M., Iriarte, X., Ros, J., and Pinto, J., 2009, "Dynamic Model of a Helical Gear Pair with Backlash and Angle-Varying Mesh Stiffness," *Multibody Dynamics, ECCOMAS Thematic Conference*, K. Arczewski, J. Fraczek and M. and Wojtyra, eds.
- [17] Siyu, C., Jinyuan, T., Caiwang, L., 2011, "Nonlinear Dynamic Characteristics of Geared Rotor Bearing Systems with Dynamic Backlash and Friction," *Mechanism and Machine Theory*, 46pp. 466-478.
- [18] Nataraj, C., and Arakere, N. K., September 12-15, 1999, "Dynamic Response and Stability of a Spur Gear Pair," 1999 ASME Design Engineering Technical Conferences, Anonymous Las Vegas, Nevada, USA, DETC99/VIB-8110, pp. 1-8.
- [19] Amabili, M., and Rivola, A., 1997, "Dynamic Analysis of Spur Gear Pairs: Steady-State Response and Stability of the Sdof Model with State-varying Meshing Damping," *Mechanical Systems and Signal Processing*, 11(3) pp. 375-390.
- [20] Theodossiades, S., and Natsiavas, S., 2001, "Periodic and Chaotic Dynamics of Motor-Driven Gear-Pair Systems with Backlash," *Chaos, Solitons & Fractals*, 12(13) pp. 2427-2440.
- [21] Cai-Wan, C., 2013, "Bifurcation and Chaos of Gear Pair System Supported by Long Journal Bearings Based on Turbulent Flow Effect and Nonlinear Suspension Effect," *World Journal of Mechanics*, 3pp. 277-291.
- [22] Kahraman, A., and Singh, R., 1991, "Interactions between State-varying Mesh Stiffness and Clearance Non-Linearities in a Geared System," *Journal of Sound and Vibration*, 146(1) pp. 135-156.
- [23] Walha, L., Fakhfakh, T., and Haddar, M., 2009, "Nonlinear Dynamics of a Two-Stage Gear System with Mesh Stiffness Fluctuation, Bearing Flexibility and Backlash," *Mechanism and Machine Theory*, 44(5) pp. 1058-1069.
- [24] Lin, J., and Parker, R. G., 2002, "Mesh Stiffness Variation Instabilities in Two-Stage Gear Systems," *Journal of Vibration and Acoustics*, 124(1) pp. 68-76.
- [25] Al-Shyyab, A. S., 2003, "Non-Linear Dynamic Analysis of a Multi-Mesh Gear Train using Multi-Term Harmonic Balance Method."
- [26] Tordion, G., and Gauvin, R., 1977, "Dynamic Stability of a Two-Stage Gear Train Under the Influence of Variable

Meshing Stiffnesses," *Journal of Manufacturing Science and Engineering*, 99(3) pp. 785-791.

[27] Parker, R., Vijayakar, S., and Imajo, T., 2000, "Non-Linear Dynamic Response of a Spur Gear Pair: Modelling and Experimental Comparisons," *Journal of Sound and Vibration*, 237(3) pp. 435-455.

[28] Sato, K., and Yamamoto, S., 1991, "Bifurcation Sets and Chaotic States of a Gear System Subjected to Harmonic Excitation," *Computational Mechanics*, 7(3) pp. 173-182.

[29] Blankenship, G. W., and Singh, R., 1995, "Analytical Solution for Modulation Sidebands Associated with a Class of Mechanical Oscillators," *Journal of Sound and Vibration*, 179(1) pp. 13-36.

[30] Kahraman, A., and Blankenship, G., 1996, "Interactions between Commensurate Parametric and Forcing Excitations in a System with Clearance," *Journal of Sound and Vibration*, 194(3) pp. 317-336.

[31] Blankenship, G., and Kahraman, A., 1995, "Steady State Forced Response of a Mechanical Oscillator with Combined Parametric Excitation and Clearance Type Non-Linearity," *Journal of Sound and Vibration*, 185(5) pp. 743-765.

[32] Kaplan, J., Dousti, S., Allaire, P., Nichols, B., Dimond, T., and Untaroiu, A., 2013, "Rotor Dynamic Modeling of Gears and Geared Systems," *ASME Proceedings*, GT2013-94654.

[33] Chaudhry, J. A., 2008., "Rotor Dynamic Analysis in MATLAB Framework," MS thesis. University of Virginia.

[34] Stringer, D.B, Sheth, P.N. 2009. "Geared Rotor Dynamic Methodologies for Advancing Prognostic Modeling Capabilities in Rotary-Wing Transmission Systems." PhD Dissertation. University of Virginia.

[35] Spotts, M.F, 1985, *Design of Machine Elements*, 6th Edition, Englewood Cliffs, NJ: Prentice-Hall, Inc.

ANNEX A

GEAR MESH COORDINATE TRANSFORMATION

$$K_{ii} = \begin{bmatrix} (s\varphi c\phi_x + c\varphi c\phi_y)^2 & & & & \\ (s\varphi c\phi_x + c\varphi c\phi_y)(s\varphi c\phi_y - c\varphi c\phi_x) & (s\varphi c\phi_y - c\varphi c\phi_x)^2 & & & \\ c\phi_z(s\varphi c\phi_x + c\varphi c\phi_y) & c\phi_z(s\varphi c\phi_y - c\varphi c\phi_x) & c\phi_z^2 & & \\ s\varphi c\phi_z r_i(s\varphi c\phi_x + c\varphi c\phi_y) & s\varphi c\phi_z r_i(s\varphi c\phi_y - c\varphi c\phi_x) & s\varphi c\phi_z^2 r_i & (s\varphi c\phi_z r_i)^2 & \\ -c\varphi c\phi_z r_i(s\varphi c\phi_x + c\varphi c\phi_y) & -c\varphi c\phi_z r_i(s\varphi c\phi_y - c\varphi c\phi_x) & -c\varphi c\phi_z^2 r_i & -c\varphi s\varphi(c\phi_z r_i)^2 & (c\varphi c\phi_z r_i)^2 \\ -c\phi_x r_i(s\varphi c\phi_x + c\varphi c\phi_y) & -c\phi_x r_i(s\varphi c\phi_y - c\varphi c\phi_x) & -c\phi_x c\phi_z r_i & -s\varphi c\phi_x c\phi_z r_i^2 & c\varphi c\phi_x c\phi_z r_i^2 & (c\phi_x r_i)^2 \end{bmatrix} \quad \text{sym}$$

$$K_{ji} = \begin{bmatrix} -(s\varphi c\phi_x + c\varphi c\phi_y)^2 & [K_{ji}]_{2,1} & [K_{ji}]_{3,1} & -s\varphi c\phi_z r_i(s\varphi c\phi_x + c\varphi c\phi_y) & c\varphi c\phi_z r_i(s\varphi c\phi_x + c\varphi c\phi_y) & c\phi_x r_i(s\varphi c\phi_x + c\varphi c\phi_y) \\ -(s\varphi c\phi_x + c\varphi c\phi_y)(s\varphi c\phi_y - c\varphi c\phi_x) & -(s\varphi c\phi_y - c\varphi c\phi_x)^2 & [K_{ji}]_{3,2} & -s\varphi c\phi_z r_i(s\varphi c\phi_y - c\varphi c\phi_x) & c\varphi c\phi_z r_i(s\varphi c\phi_y - c\varphi c\phi_x) & c\phi_x r_i(s\varphi c\phi_y - c\varphi c\phi_x) \\ -c\phi_z(s\varphi c\phi_x + c\varphi c\phi_y) & -c\phi_z(s\varphi c\phi_y - c\varphi c\phi_x) & -c\phi_z^2 & -s\varphi c\phi_z^2 r_i & c\varphi c\phi_z^2 r_i & c\phi_x c\phi_z r_i \\ -s\varphi c\phi_z r_j(s\varphi c\phi_x + c\varphi c\phi_y) & -s\varphi c\phi_z r_j(s\varphi c\phi_y - c\varphi c\phi_x) & -s\varphi c\phi_z^2 r_j & -(s\varphi c\phi_z)^2 r_i r_j & c\varphi s\varphi c\phi_z^2 r_i r_j & s\varphi c\phi_x c\phi_z r_i r_j \\ c\varphi c\phi_z r_j(s\varphi c\phi_x + c\varphi c\phi_y) & c\varphi c\phi_z r_j(s\varphi c\phi_y - c\varphi c\phi_x) & c\varphi c\phi_z^2 r_j & c\varphi s\varphi c\phi_z^2 r_i r_j & -(c\varphi c\phi_z)^2 r_i r_j & -c\varphi c\phi_x c\phi_z r_i r_j \\ -c\phi_x r_j(s\varphi c\phi_x + c\varphi c\phi_y) & -c\phi_x r_j(s\varphi c\phi_y - c\varphi c\phi_x) & -c\phi_x c\phi_z r_j & -s\varphi c\phi_x c\phi_z r_i r_j & c\varphi c\phi_x c\phi_z r_i r_j & c\phi_x^2 r_i r_j \end{bmatrix}$$

$$K_{jj} = \begin{bmatrix} (s\varphi c\phi_x + c\varphi c\phi_y)^2 & & & & \\ (s\varphi c\phi_x + c\varphi c\phi_y)(s\varphi c\phi_y - c\varphi c\phi_x) & (s\varphi c\phi_y - c\varphi c\phi_x)^2 & & & \\ c\phi_z(s\varphi c\phi_x + c\varphi c\phi_y) & c\phi_z(s\varphi c\phi_y - c\varphi c\phi_x) & c\phi_z^2 & & \\ s\varphi c\phi_z r_j(s\varphi c\phi_x + c\varphi c\phi_y) & s\varphi c\phi_z r_j(s\varphi c\phi_y - c\varphi c\phi_x) & s\varphi c\phi_z^2 r_j & (s\varphi c\phi_z r_j)^2 & \\ -c\varphi c\phi_z r_j(s\varphi c\phi_x + c\varphi c\phi_y) & -c\varphi c\phi_z r_j(s\varphi c\phi_y - c\varphi c\phi_x) & -c\varphi c\phi_z^2 r_j & -c\varphi s\varphi(c\phi_z r_j)^2 & (c\varphi c\phi_z r_j)^2 \\ c\phi_x r_j(s\varphi c\phi_x + c\varphi c\phi_y) & c\phi_x r_j(s\varphi c\phi_y - c\varphi c\phi_x) & c\phi_x c\phi_z r_j & s\varphi c\phi_x c\phi_z r_j^2 & -c\varphi c\phi_x c\phi_z r_j^2 & (c\phi_x r_j)^2 \end{bmatrix} \quad \text{sym}$$

(16)